

Futures convexity adjustment with continuously and discretely compounded rates

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Abstract

The paper deals with interest rate futures on interbank fixings. With analytical results we quantify and compare the difference between futures convexity adjustment for continuously and discretely compounded forward rates, respectively. We find that the use of continuously compounded rates can underestimate the futures convexity adjustment by more than the typical bid-offer spread in a liquid market. Analytical formulas for futures convexity adjustment are derived. The Ho-Lee model is used as the calculation framework.

1 Introduction

Calculations with continuously compounded rates are often easier than the discretely compounded counterparts. The paper has the simple mission of clarifying whether discretely compounded rates are necessary to take into account when calculating the futures convexity adjustment. Calculations are carried out with the Ho-Lee model.

The paper is structured as follows. Section 2 make the foundation for calculating the expressions for the convexity adjustments. The convexity adjustment for futures based on continuously compounded rate is done in section 3. The convexity adjustment for futures based on discretely compounded rate is done in section 4. Numerical results are discussed in section 5. Finally we conclude in section 6.

2 Preliminaries

2.1 Futures

Lemma 2.1. *Let $\Phi(t)$ be any process. Let T be given. If $\Phi(t) = \mathbb{E}_t[\Phi(T)] \forall t \leq T$ then for any s such that $t \leq s \leq T$:*

$$\Phi(t) = \mathbb{E}_t[\Phi(s)]$$

Proof: By rule of iterated expectations:

$$\begin{aligned}\Phi(t) &= \mathbb{E}_t[\Phi(T)] \\ &= \mathbb{E}_t[\mathbb{E}_s[\Phi(T)]] \\ &= \mathbb{E}_t[\Phi(s)]\end{aligned}$$

◻

Note If $\Phi(t) = \mathbb{E}_t[\Phi(T)]$ for all t and T such that $t \leq T$ then $\Phi(t)$ is a martingale.

Proposition 2.2. *Let $\Phi(t)$ be a futures process such that $\Phi(t) = \Psi(T)$ at $t = T$. Then $\Phi(t)$ is a Q -martingale, and*

$$\Phi(t) = \mathbb{E}_t^Q[\Psi(T)]$$

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Note $\Psi(T)$ is some stochastic variable known at time T . Here it is relevant to think of $\Psi(T)$ as the Libor fixing fixed at time T , $F(T, T, T + \Delta T)$.

Proof: (Arguments copied from Andersen and Piterbarg (2010). An earlier reference is Hunt and Kennedy (2000).) Since the value of a futures contract always is zero then

$$\mathbb{E}_t^Q \left[\int_t^\tau \frac{1}{B(t, s)} d\Phi(s) \right] = 0$$

where

$$dB(t, s) = B(t, s)r(s)ds \text{ and } B(t, t) = 1$$

This holds for all t and τ such that $t \leq \tau \leq T$ and since $B(t, s)$ bounded and positive then

$$\mathbb{E}_t^Q[d\Phi(s)] = 0 \quad \forall s$$

and therefore $\Phi(t)$ is a martingale. The expectation follows directly from the fact that $\Phi(t)$ is a martingale. $\circ$

2.2 Convexity Adjustment Definition

We define the convexity adjustment as the difference between the market quote K of the future, and the forward rate $F(t, T_1, T_2)$ at T_1 :

$$CA(t, T_1) = K(t, T_1) - F(t, T_1, T_2) \quad (1)$$

2.3 Girsanov

Theorem 2.3. Girsanov's. *If Q_1 and Q_2 are equivalent measures then there exists a ϕ (the Girsanov kernel) such that*

$$dW^{Q_1} = \phi(t)dt + dW^{Q_2} \quad (2)$$

Proof. See Bjork (2009), prop 11.3. $\square$

Proposition 2.4. *Assume numeraire S_i for Q_i and $dS_i = \dots dt + \sigma_i(t)S_i(t)dW^{Q_i}$ where σ_i is adapted to W^{Q_i} for $i \in \{1, 2\}$. Then the Girsanov kernel for the transition from Q_1 to Q_2 as defined by (2) is given by*

$$\phi(t) = \sigma_2(t) - \sigma_1(t)$$

Proof. See Bjork (2009), prop 26.5. $\square$

Corollary 2.5. Going from Q to Q^T *From theorem 2.3 and prop. 2.4 with $S_1(t) = B(t)$ and $S_2(t) = p(t, T)$ with the dynamics $dp(t, T) = \dots dt + \sigma_p(t, T)p(t, T)dW$ and $dB(t) = r(t)B(t)dt$ then*

$$dW^Q = \sigma_p(t, T)dt + dW^{Q^T} \quad (3)$$

2.4 Instantaneous forward rate

The instantaneous forward rates on the discount curve can be defined from

$$p(t, T) = \exp\left(-\int_t^T f(t, s)ds\right)$$

It then follows

$$f(t, T) = -\frac{1}{p(t, T)} \frac{\delta p(t, T)}{\delta T}$$

Alternatively define $f(t, T)$ as the instantaneous rate at time T set at time t . $f(t, T)$ can then be derived from the replication argument of buying $p(t, T + \Delta T)/p(t, T)$ number of $p(t, T)$ and selling one $p(t, T + \Delta T)$. The price at time t of this portfolio is 0. At time T we receive $p(t, T + \Delta T)/p(t, T)$ and at time $T + \Delta T$ we pay one. Taking $\Delta T \rightarrow 0$ this implies a forward rate of

$$f(t, T) = -\frac{1}{p(t, T)} \frac{\delta p(t, T)}{\delta T}$$

2.5 Bank account

Define the bank account as

$$B(t, T) = \exp\left(\int_t^T r(s) ds\right)$$

where r is the short rate and $r(t) = f(t, t)$. From risk neutral evaluation

$$p(t, T) = \mathbb{E}_t^Q[1/B(t, T)]$$

2.6 The forward rate is the expected short rate

At time t then $f(T, T)$ is a stochastic variable. The price at time t of the pay off $f(T, T) - K$ is given by

$$\begin{aligned} \text{PV} &= \mathbb{E}_t^Q\left[\frac{1}{B(t, T)}(f(T, T) - K)\right] \\ &= p(t, T)(\mathbb{E}_t^{Q^T}[f(T, T)] - K) \\ \text{PV} &= 0 \Rightarrow K = \mathbb{E}_t^{Q^T}[f(T, T)] \end{aligned}$$

I.e. at price=0 we can at time t borrow at time T at a rate equal to $\mathbb{E}_t^{Q^T}[f(T, T)]$. Cf. section 2.4 this rate was defined as $f(t, T)$. I.e.

$$\begin{aligned} f(t, T) &= \mathbb{E}_t^{Q^T}[f(T, T)] \\ &= \mathbb{E}_t^{Q^T}[r(T)] \end{aligned} \tag{4}$$

2.7 f Dynamics under Q^T

From (4) and lemma 2.1 it is evident that $f(\cdot, T)$ is a martingale under Q^T and

$$df(t, T) = \sigma_f(t, T)dW^{Q^T}(t) \tag{5}$$

where $\sigma_f(t, T)$ is adapted to W^{Q^T} .

2.8 p Dynamics under Q

Under Q we know that the drift of the bond is $r(t)$:

$$dp(t, T) = r(t)p(t, T)dt + \sigma_p(t, T)p(t, T)dW^Q \tag{6}$$

$$\text{where} \quad \sigma_p(t, T) \equiv - \int_t^T \sigma_f(t, s) ds \tag{7}$$

The expression for $\sigma_p(t, T)$ in (7) is found as follows. From Ito and (6):

$$d \log p(t, T) = \dots dt + \sigma_p(t, T)dW$$

Since $\log p(t, T) = - \int_t^T f(t, u)du$ then integrate both sides of (5) wrt T and compare:

$$\begin{aligned} d \log p(t, T) &= d\left(- \int_t^T f(t, u)du\right) \\ &\stackrel{\forall t_0}{=} d(f(t_0, u) + \int_t^T \int_{t_0}^t \sigma_f(s, u) dW^{Q^T} du) \\ &= d(f(t_0, u) + \int_{t_0}^t \int_t^T \sigma_f(s, u) du dW^{Q^T}) \\ &= \int_t^T \sigma_f(s, u) du dW^{Q^T} \end{aligned}$$

2.9 Ho-Lee model

The Ho-Lee model short rate dynamics is given by

$$dr = \theta(t)dt + \sigma dW^Q \quad (8)$$

where θ is deterministic and fitted to match the current forward rate structure and σ is constant.

Note that with $\sigma_f(t, T) = \sigma$ then the dynamics of the forward rates imply the short rate Ho-Lee model, see e.g. Bjork (2009).

3 Convexity adjustment of a continuously compounded forward rate

The derivations for the deriving the convexity adjustment for continuously compounded forward rates are highly inspired by Hull.

According to (6) then for the Ho-Lee model (under Q)

$$dp(t, T) = r(t)p(t, T)dt - (T - t)\sigma p(t, T)dW^Q(t)$$

The continuously compounded rate can be stated as:

$$f(t, T_1, T_2) = \frac{\log(p(t, T_1) - \log(p(t, T_2)))}{T_{12}} \quad (9)$$

where $T_{12} = T_2 - T_1$. Itô's lemma applied to (9):

$$df(t, T_1, T_2) = \frac{\sigma^2(T_2 - t)^2 - \sigma^2(T_1 - t)^2}{2T_{12}}dt - \sigma dW^Q(t) \quad (10)$$

Taking expectation under Q at time $t = T_1$ and using $\frac{1}{T_{12}} \int_0^{T_1} (T_2 - t)^2 - (T_1 - t)^2 dt = T_1 T_2$:

$$E^Q[f(T_1, T_1, T_2)] = f(0, T_1, T_2) + \frac{1}{2}\sigma^2 T_1 T_2$$

I.e.

$$CA_C = \frac{1}{2}\sigma^2 T_1 T_2 \quad (11)$$

4 Convexity adjustment of discretely compounded forward rate

The simple forward rate is expressed by the continuously compounded forward rate:

$$F(t, T_1, T_2) = \frac{1}{\delta_{12}}(\exp(f(t, T_1, T_2)T_{12}) - 1) \quad (12)$$

where δ_{12} is the coverage for the the discretely compounded rates, ie the standard money market basis, which is act/360 for the main currencies. T_{12} is the coverage for the continuously compounded rates, which can have any basis.

Itô's lemma on (12), where $f = f(t, T_1, T_2)$:

$$\begin{aligned} dF(t, T_1, T_2) &= \frac{\delta F}{\delta f} df + \frac{1}{2} \frac{\delta^2 F}{\delta^2 f} df^2 \\ &= \frac{T_{12}}{\delta_{12}} \exp(f(t, T_1, T_2)T_{12}) df + \frac{1}{2} \frac{T_{12}^2}{\delta_{12}} \exp(f(t, T_1, T_2)T_{12}) df^2 \\ &\stackrel{(10)}{=} \frac{1}{2} \sigma^2 \left(\frac{(T_2 - t)^2 - (T_1 - t)^2}{\delta_{12}} + \frac{T_{12}^2}{\delta_{12}} \right) \exp(f(t, T_1, T_2)T_{12}) dt \\ &\quad + \sigma \frac{T_{12}}{\delta_{12}} \exp(f(t, T_1, T_2)T_{12}) dW(t) \\ &= \frac{1}{\delta_{12}} \sigma^2 (T_2^2 - tT_{12} - T_1 T_2) \exp(f(t, T_1, T_2)T_{12}) dt \\ &\quad + \sigma \frac{T_{12}}{\delta_{12}} \exp(f(t, T_1, T_2)T_{12}) dW(t) \end{aligned}$$

Define $K(f(t, T_1, T_2)) = \exp(f(t, T_1, T_2) T_{12})$ then $dK(f) = \delta_{12} dF(t, T_1, T_2)$. I.e.

$$dK(f(t, T_1, T_2)) = \sigma^2 (T_2^2 - tT_{12} - T_1T_2) K(f(t, T_1, T_2)) dt + T_{12}\sigma K(f(t, T_1, T_2)) dW(t)$$

Taking expectation of this GBM at $t = T_1$:

$$\begin{aligned} E^Q [K(f(T_1, T_1, T_2))] &= K(f(0, T_1, T_2)) \cdot \exp \left\{ \sigma^2 \int_0^{T_1} T_2^2 - tT_{12} - T_1T_2 dt \right\} \\ &= K(f(0, T_1, T_2)) \cdot \exp \left\{ \frac{1}{2} \sigma^2 T_1 T_{12} (T_1 + 2T_{12}) \right\} \end{aligned}$$

And then

$$E^Q [F(T_1, T_1, T_2)] = \exp \left\{ \frac{1}{2} \sigma^2 T_1 T_{12} (T_1 + 2T_{12}) \right\} \left(\frac{1}{\delta_{12}} + F(0, T_1, T_2) \right) - \frac{1}{\delta_{12}}$$

I.e. the convexity adjustment for the discretely compounded forward rate is given by (1)

$$CA_D = \left(F(0, T_1, T_2) + \frac{1}{\delta_{12}} \right) \left(\exp \left\{ \frac{1}{2} \sigma^2 T_1 T_{12} (T_1 + 2T_{12}) \right\} - 1 \right) \quad (13)$$

5 Numerical examples

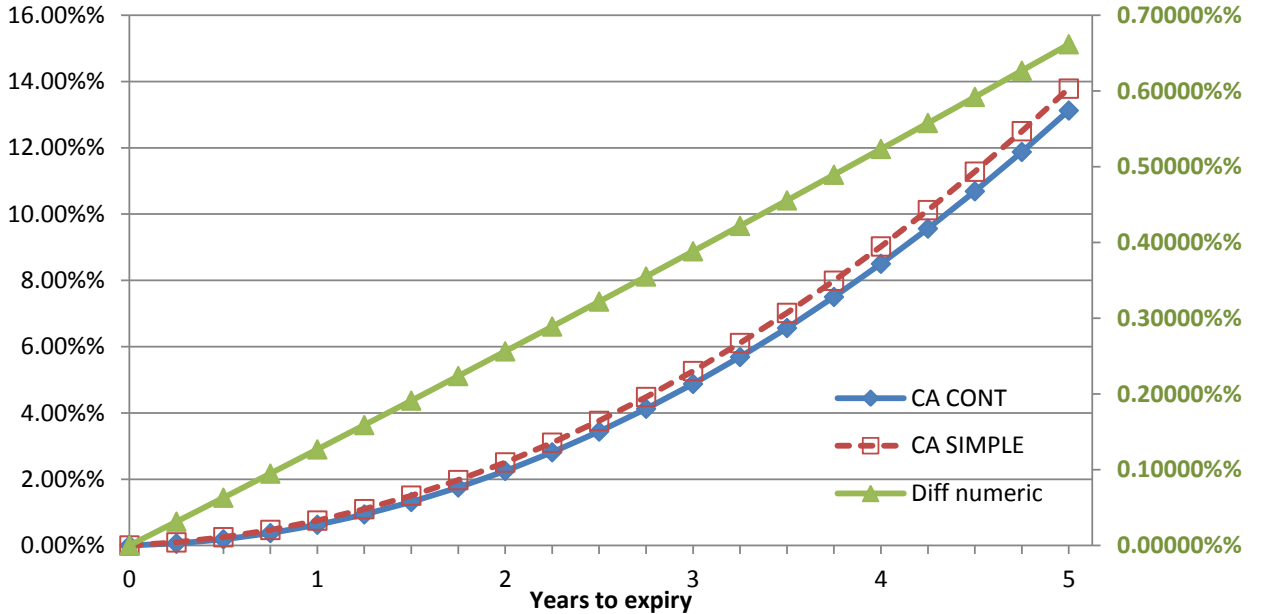


Figure 1: Convexity adjustment in basis points for continuously compounded rates (11) and discretely compounded rates (13) respectively - on primary axis. The exact difference on secondary axis. T_1 is time in years to expiry. Input parameters: $\sigma = 1\%$, $T_{12} = 0.25$, $\delta_{12} = 0.25$, $F(0, T_1, T_2) = 1\%$. $T_2 = T_1 + T_{12}$. Y-axis is basis points. X-axis is years.

In figure 1 we plot the convexity adjustments for continuously and discretely compounded rates (primary axis), and the difference (secondary axis). The length of length of the forward rates in question is 3 months, and the volatility has been chosen to $\sigma = 0.01$. For this figure both δ_{12} and T_{12} was set to 0.25, resembling 3 months. The choice of day count basis to calculate δ_{12} and T_{12} has only little effect on the convexity adjustments. The current forward rate $F(0, T_1, T_2)$ also has only little effect on the difference.

On a longer time scale had been used then we would see the difference rises fast. However, this is not of any concern with the market instruments currently traded. With $\sigma < 0.001$ then the difference stays below 0.25 bp for $T_1 < 50$.

6 Conclusion

We have qualified and quantified the difference in futures convexity adjustment for continuously and discretely compounded forward rates for the Ho-Lee model. Analytical results for the difference are given. We find that for $\sigma \gtrsim 1\%$ then, if possible, calculations should be based on discretely compounded forward rates as the convexity adjustment achieved using calculations based on continuously compounded rates underestimate the real (model) value by more than the typical bid-offer spread. using the discretely compounded result matter when

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